Detailed Note on Projection-Based Inference for the Rank-1 Matrix Model with $\log_{10}(EC) - \log_{10}(EO)$

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1 Purpose

This note documents the inference procedure used for the ROI-by-band matrix model with predictor

$$X_i = \log_{10}(EC_i) - \log_{10}(EO_i),$$

where the matrix rank is fixed at $r = 1$ according to the 5-fold cross-validation analysis. The goal is not to re-select the rank, but to perform valid uncertainty quantification for a prespecified collection of linear functionals of the fitted low-rank coefficient matrix.

The implementation is driven by the scripts

- `run_roi10_matproj_log_contrast_rank1_ci_analysis.m`,
- `matProj.m`,
- `fit_low_rank_trace_regression.m`.

2 Data and model

The input file is

`hbn_bandpower_8band_roi10.mat`,

which contains $N = 397$ subjects. For subject i, the analysis uses:

- response $y_i = \text{age}_i$;
- 10×8 eyes-closed ROI band-power matrix EC_i;
- 10×8 eyes-open ROI band-power matrix EO_i.

The matrix regression model is

$$y_i = \langle X_i, B^* \rangle + \varepsilon_i, \qquad \operatorname{rank}(B^*) \leq 1.$$

There is no explicit intercept term, so centering is handled inside the inference routine.

3 Construction of the log-contrast matrix

Let EC_i and EO_i denote the raw ROI power matrices in μV^2. Before taking logarithms, the code defines a numerical floor

$$\text{floor} = \max\{0.5 \times \min(\text{positive raw EO/EC entries}),\ 10^{-12}\}.$$

The transformed matrices are then

$$\widetilde{E}_i^{EC} = \log_{10}(\max(EC_i, \text{floor})), \qquad \widetilde{E}_i^{EO} = \log_{10}(\max(EO_i, \text{floor})),$$

and the final matrix covariate is

$$X_i = \widetilde{E}_i^{EC} - \widetilde{E}_i^{EO}.$$

For the current data set, the numerical floor used in this analysis is

$$2.15735373786 \times 10^{-4}.$$

4 Inference targets

The inferential targets are linear functionals of the form

$$\Delta_T(B^*) = \langle T, B^* \rangle,$$

where T is a user-specified loading matrix. The current analysis evaluates the following pre-specified loadings:

- **single-element loadings**: one loading for each ROI-band entry, yielding $10 \times 8 = 80$ targets;
- **ROI-slice loadings**: one loading per ROI, with ones across all eight bands, yielding 10 targets;
- **band-slice loadings**: one loading per band, with ones across all ten ROIs, yielding 8 targets;
- **global loading**: the 10×8 all-ones matrix, yielding 1 target.

Every loading is normalized to have unit Frobenius norm before inference.

5 Cross-fitted inference algorithm

For a fixed loading T, the function `matProj(X, y, T, 1)` performs the following steps.

Step 1: random half-splitting

The $N = 397$ subjects are split into two folds with sizes

$$n_1 = \lfloor N/2 \rfloor = 198, \qquad n_2 = N - n_1 = 199.$$

The loading-analysis script resets the random number generator to the common seed

$$20260321$$

before every call to `matProj`, so all loadings are evaluated on the same sample split.

Step 2: fold-specific centering

Within each half, the matrix predictors and response are centered:

$$X_i^{(c)} = X_i - \bar{X}, \qquad y_i^{(c)} = y_i - \bar{y}.$$

The stored fold means are later reused to center the opposite evaluation half. Hence the current implementation uses fold-specific centering throughout the initialization, residual construction, and variance calculation.

Step 3: foldwise second moments

For each training half, the routine computes

- the mode-1 second-moment matrix S_1,
- the mode-2 second-moment matrix S_2,
- the average squared Frobenius norm $\overline{\|X\|_F^2}$.

These are built from the centered training-half matrices and enter the covariance-weighted projection formulas.

Step 4: low-rank initialization

On each training half, the initial estimator is obtained by fitting a rank-1 low-rank trace regression to the centered data:

$$\min_{\operatorname{rank}(B)\le 1} \frac{1}{2n}\sum_{i=1}^{n}\big(y_i^{(c)} - \langle X_i^{(c)}, B\rangle\big)^2.$$

The numerical solver is projected gradient descent with truncated SVD projection:

1. reshape the matrices into a design matrix Z;
2. compute a step size from the Lipschitz constant $L = \sigma_{\max}(Z)^2/n$;
3. take a gradient step;
4. project back to rank 1 by truncated SVD;
5. stop when the relative coefficient change or the relative objective change falls below 10^{-7}, or after 1000 iterations.

The output is the rank-1 matrix initializer and its left/right singular vectors.

Step 5: opposite-fold evaluation

Given a training half and an evaluation half, the evaluation matrices are centered using the *training-half* means. The method then computes:

- the residuals from the training-half initializer;
- the empirical score-like matrix E_{mean};
- the projection direction $\Phi(T)$ through `phi_sigma_mat`;
- the projected plug-in term through `P_sigma_mat`.

This yields one half-specific estimator of $\Delta_T(B^*)$.

Step 6: cross-fitted point estimator

The final point estimator averages the two half-specific estimators:

$$\widehat{\Delta}_T = \frac{\widehat{\Delta}_T^{(1\leftarrow 2)} + \widehat{\Delta}_T^{(2\leftarrow 1)}}{2}.$$

Here the notation indicates that the first superscript half is used for evaluation and the second for training.

Step 7: standard error and confidence interval

The residuals from both evaluation halves are pooled to estimate σ^2. The routine then forms the quadratic variance term

$$\widehat{\Phi} = \frac{1}{2}\Big\{\Phi_1^\top\big[\mathrm{kron}(S_{2,1}^{eval}/\overline{\|X\|_{F,1}^2},\, S_{1,1}^{eval})\big]\Phi_1 + \Phi_2^\top\big[\mathrm{kron}(S_{2,2}^{eval}/\overline{\|X\|_{F,2}^2},\, S_{1,2}^{eval})\big]\Phi_2\Big\},$$

and reports

$$\widehat{\mathrm{sd}}_T = \sqrt{\widehat{\sigma}^2\,\widehat{\Phi}}.$$

The nominal 95% confidence interval is

$$\widehat{\Delta}_T \pm 1.96\frac{\widehat{\mathrm{sd}}_T}{\sqrt{N}}.$$

6 Current output summary

The current run evaluates 99 loadings in total:

$$80 \text{ single-element} + 10 \text{ ROI-slice} + 8 \text{ band-slice} + 1 \text{ global}.$$

The significance summary is:

- 6/80 single-element loadings exclude zero;
- 2/10 ROI-slice loadings exclude zero;
- 1/8 band-slice loadings exclude zero;
- the global loading does not exclude zero.

The most important structured results are:

- anterior-midline ROI-slice: $\hat{\delta} = 0.562$, 95% CI [0.169, 0.956];
- central-right ROI-slice: $\hat{\delta} = -0.529$, 95% CI [−0.979, −0.080];
- alpha2 band-slice: $\hat{\delta} = 0.715$, 95% CI [0.243, 1.187];
- global loading: $\hat{\delta} = 0.045$, 95% CI [−0.146, 0.237].

Thus the inference points to localized structure rather than a uniform matrix-wide shift.

7 Files generated by the analysis

The rank-1 matrix inference results are written to:

- `matproj_log_contrast_rank1_all_results.csv`,
- `matproj_log_contrast_rank1_summary.csv`,
- `log_contrast_rank1_single_element_heatmap.png`,
- `log_contrast_rank1_structured_loadings.png`,
- `matproj_log_contrast_rank1_summary.txt`,
- `matproj_log_contrast_rank1_config.json`.

Figure 1 reproduces the structured-loading plot written by the script.

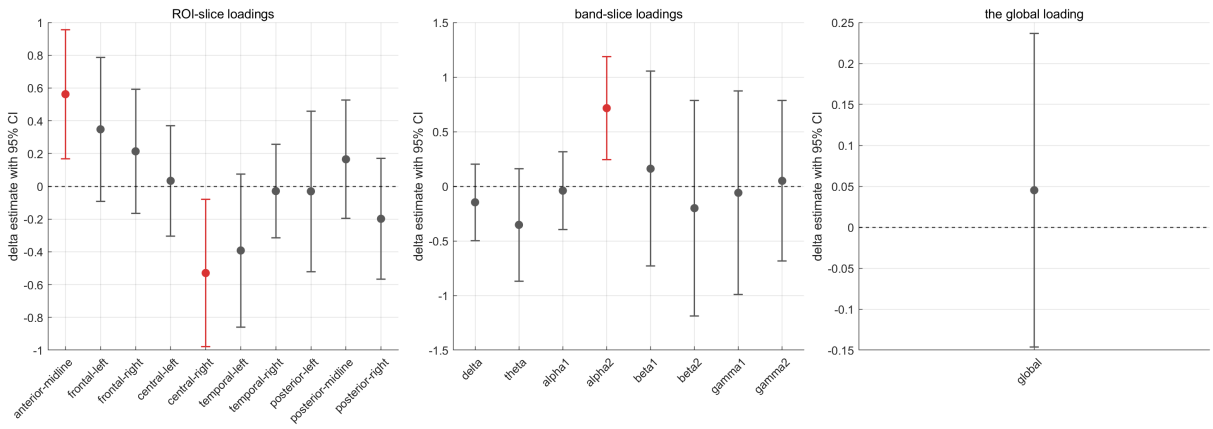

Figure 1: Projection-based confidence intervals for the ROI-slice, band-slice, and global loadings under the rank-1 log-contrast model.

8 Interpretation

From a methodological perspective, this analysis illustrates the role of the proposed projection-based inference framework after rank selection. The rank-1 model is determined by cross-validation, but the scientific conclusions come from the confidence intervals attached to pre-specified structured loadings. In the present application, the fitted low-rank log-contrast signal is supported mainly through the anterior-midline and central-right ROI summaries and the alpha2 band summary, whereas the global loading is not distinguishable from zero.